IE4728 Enterprise Network Design Project

Network Design Review
(Week 6 and 7)

Review of

1. IP Addressing and Subnetting
2. CIDR and Supernetting
3. Optimization of Network Design
4. Access Control List

Review of
IP Addressing and Subnetting

IP Address Classes

	1 Byte ← 8 Bits →	1 Byte ← 8 Bits →	1 Byte ← 8 Bits →	1 Byte ← 8 Bits →	Do not assign network addresses:
0000 0000: 0 0111 1111: 127 **Class A:**	N 0 - 127	H	H	H	0.x.x.x or 127.x.x.x
1000 0000: 128 1011 1111: 191 **Class B:**	N 128 - 191	N	H	H	128.0.x.x or 191.255.x.x
1100 0000: 192 1101 1111: 223 **Class C:**	N 192 - 223	N	N	H	192.0.0.x or 223.255.255.x

N = Network number assigned externally by ICANN
H = Host number assigned by network administrator

© Cisco Systems, Inc. 1999

Special IP Addresses

Prefix (Network)	Suffix (Host)	Type of Address	Purpose
All 0s x.0.0.0/16 x.0.0.0/24	All 0s	This host	Used during bootstrap
network	All 0s	network	Identifies a (this) network
network	All 1s	Directed broadcast	Broadcast on (specified) distance net
All 1s x.255.0.0/16 x.255.0.0/24	All 1s	Limited broadcast	Broadcast on (this) local net
127	any	loopback	testing

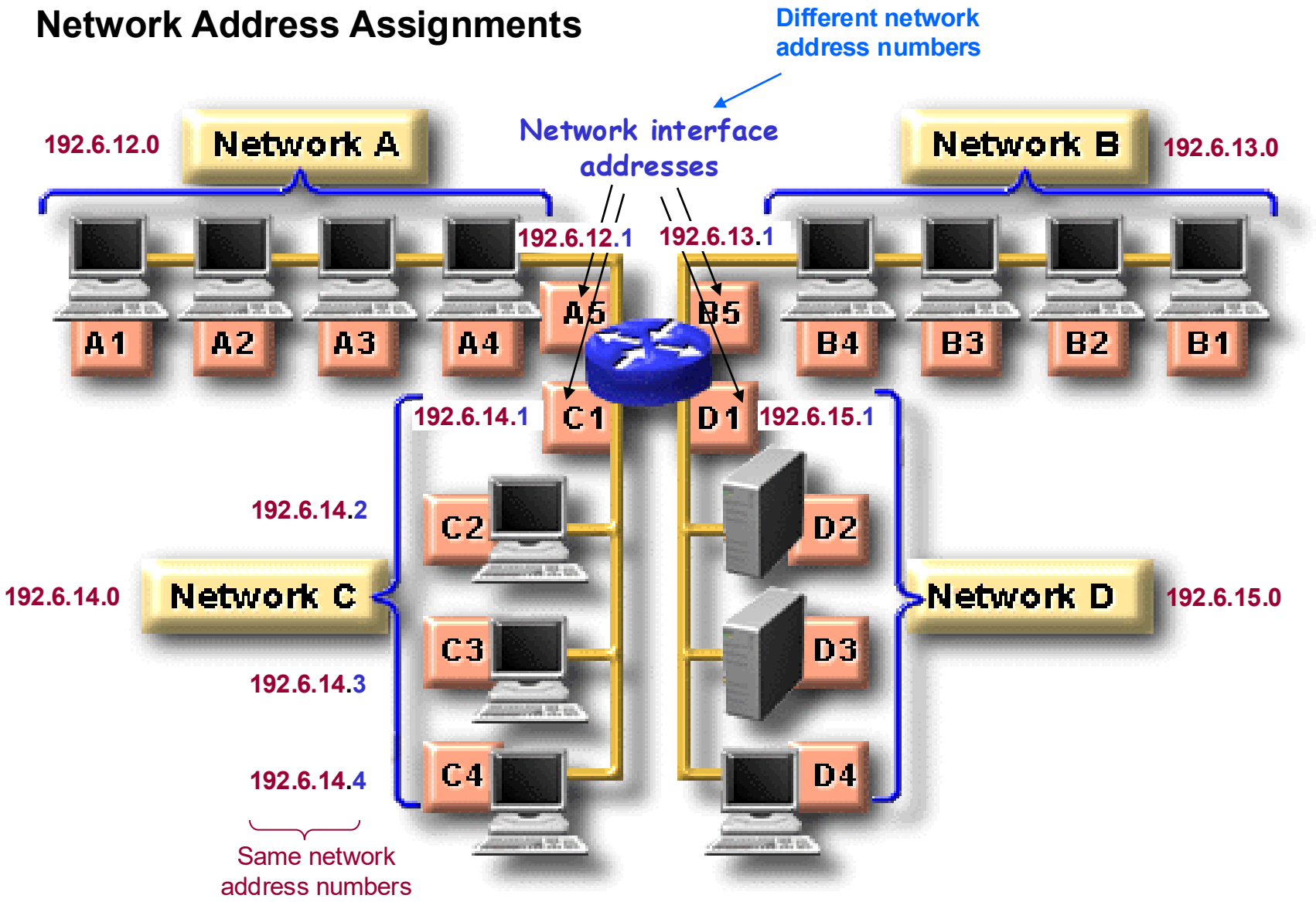
Do not assign all 0s and all 1s for network number and host number

Assigning Network Addresses

To assign network addresses, follow these rules:

- You must have a network address for each logical network.
- If your system is attached to more than one physical network via a gateway / router, the network addresses of these interfaces may not be the same.
  - All nodes in the same network, however, must have the same network address.
- Do not assign network addresses which are reserved *(all 0's or 1's)* or of special meaning *(127.0.0.0)*

Network Address Assignments

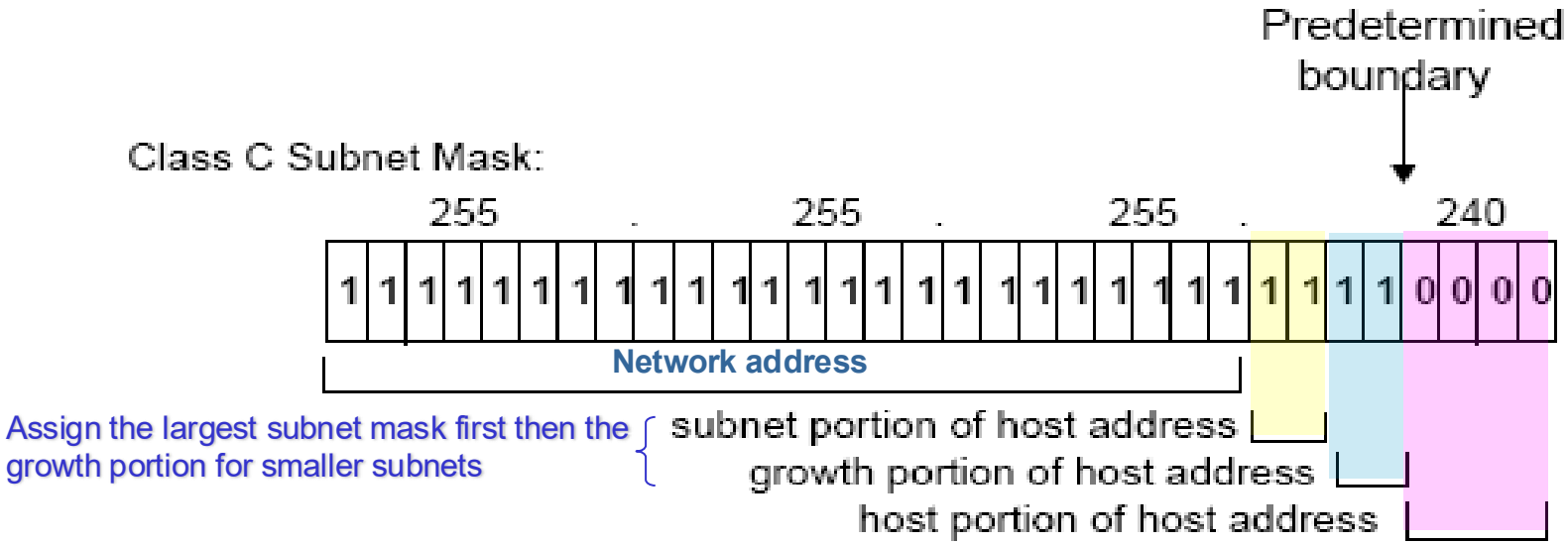


Subnetting using Variable Length Subnet Mask (VLSM)

Subnetting Strategies

- Physical Network
  - LAN segmentation produces smaller subnets to conserve network address usage
  - LAN augmentation produces larger subnets to minimize network address wastage
  - Where possible and within operational limits of cable length, local area networks should be appropriately segmented.
- Logical Network
  - Subnetting the growth bits from the MSB to LSB should be followed to efficiently assign the available network addresses
  - Supernetting where possible should be carried out to maximize route aggregation for improved router performance by filtering network addresses with common MSB bits.

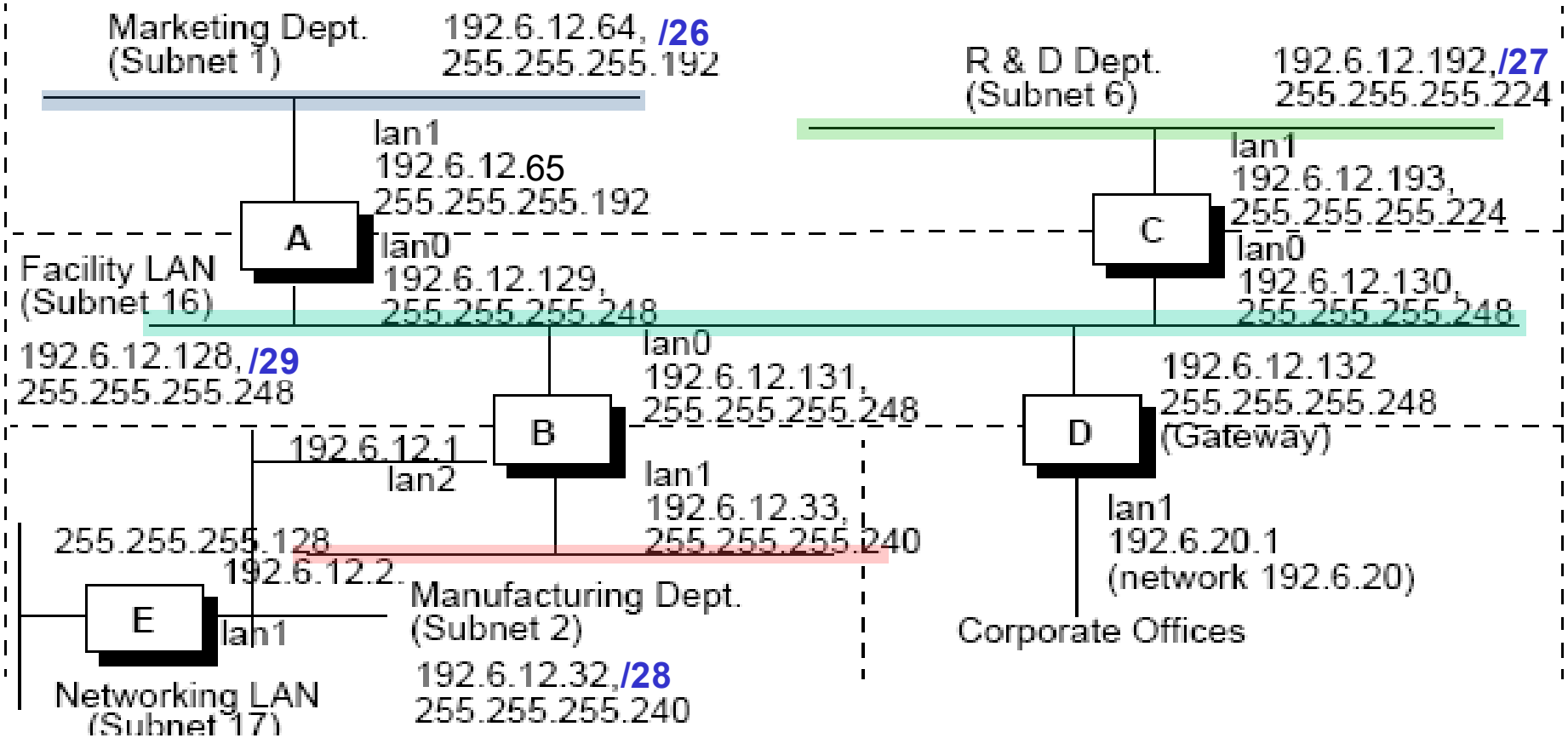
Variable-Length Subnet Addressing



Subnet	Host Address (Bits)	Subnet Mask (Bits)
A	10gg gghh	1111 1100
B	01gg ghhh	1111 1000
C	110g ghhh	1111 1000
D	001g ghhh	1111 1000

Network Map with Variable-Length Subnets

Company division network = 192.6.12.0



VLSM – Simple Example

	1st octet	2nd octet	3rd octet	4th octet
10.0.0.0/8	10	Host	Host	Host
10.0.0.0/16	10	Subnet	Host	Host
10.0.0.0/16	10	0	Host	Host
10.1.0.0/16	10	1	Host	Host
10.2.0.0/16	10	2	Host	Host
10.n.0.0/16	10	…	Host	Host
10.255.0.0/16	10	255	Host	Host

- Subnetting a /8 subnet using a /16 mask gives us 256 subnets with 65,536 hosts per subnet (excluding special subnet/host addresses is fewer by 2)
- Let's take the 10.2.0.0/16 subnet and subnet it further…

VLSM – Simple Example

	Network	Subnet	Host	Host
10.2.0.0/16	10	2	Host	Host
10.2.0.0/24	10	2	Subnet	Host
10.2.0.0/24	10	2	0	Host
10.2.1.0/24	10	2	1	Host
10.2.n.0/24	10	2	…	Host
10.2.255.0/24	10	2	255	Host

- Note: 10.2.0.0/16 is now a summary of all of the 10.2.0.0/24 subnets.

VLSM – Simple Example

10.0.0.0/8 "subnetted using /16"

255 /16 subnets

Subnet	1st host	Last host	Broadcast
10.0.0.0/16	**10.0.0.1**	**10.0.255.254**	**10.0.255.255**
10.1.0.0/16	**10.1.0.1**	**10.1.255.254**	**10.1.255.255**

10.2.0.0/16 → further "sub-subnetted using /24"

256 /24 subnets

Subnet	1st host	Last host	Broadcast
10.2.0.0/24	**10.2.0.1**	**10.2.0.254**	**10.2.0.255**
10.2.1.0/24	**10.2.1.1**	**10.2.1.254**	**10.2.1.255**
10.2.2.0/24	**10.2.2.1**	**10.2.2.254**	**10.2.2.255**
…			
10.2.255.0/24	**10.2.255.1**	**10.2.255.254**	**10.2.255.255**

10.3.0.0/16	**10.3.0.1**	**10.3.255.254**	**10.0.255.255**
…			
10.255.0.0/16	**10.255.0.1**	**10.255.255.254**	**10.255.255.255**

Your network can now have 255 /16 subnets with 65,536 hosts each
AND 256 /24 subnets with 256 hosts each

VLSM – Practical Example

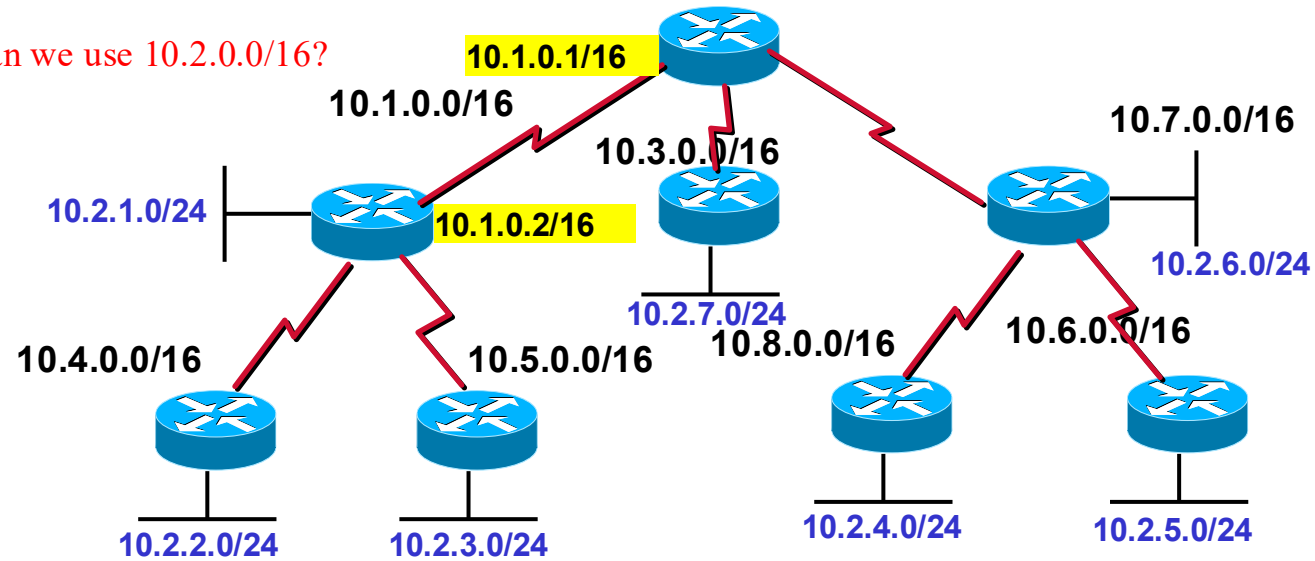
Subnets

~~10.0.0.0/16~~
10.1.0.0/16
10.2.0.0/16
  10.2.0.0/24
  10.2.1.0/24
  10.2.2.0/24
  …
  10.2.255.0/24
10.3.0.0/16
…
~~10.255.0.0/16~~

Can we use 10.2.0.0/16?

An example of VLSM, NOT a good network design

- wasteful network and host address space on serial links



- **Your network can now have 253/16 subnets with 65,534 hosts each *AND* 256 /24 subnets with 254 hosts each**

  How many subnets do we actually need?
- All you need to make it works is a classless routing protocol that passes the *subnet mask* with the *network address* in the routing updates.
- Classless routing protocols: RIPv2, EIGRP, OSPF, IS-IS, BGP

Another VLSM Example using /30 subnets

Subnetting a Class C network 207.21.24.0

207.21.24.0/24 network subnetted into eight /27 subnets

Subnet 0	207.21.24.0	/27
Subnet 1	207.21.24.32	/27
Subnet 2	207.21.24.64	/27
Subnet 3	207.21.24.96	/27
Subnet 4	207.21.24.128	/27
Subnet 5	207.21.24.160	/27
~~Subnet 6~~	~~207.21.24.192~~	~~/27~~
Subnet 7	207.21.24.224	/27

Hosts of Sub-subnet 0 /30

110|000|00 192 unusable
110|000|01 193 ok
110|000|10 194 ok
110|000|11 195 unusable

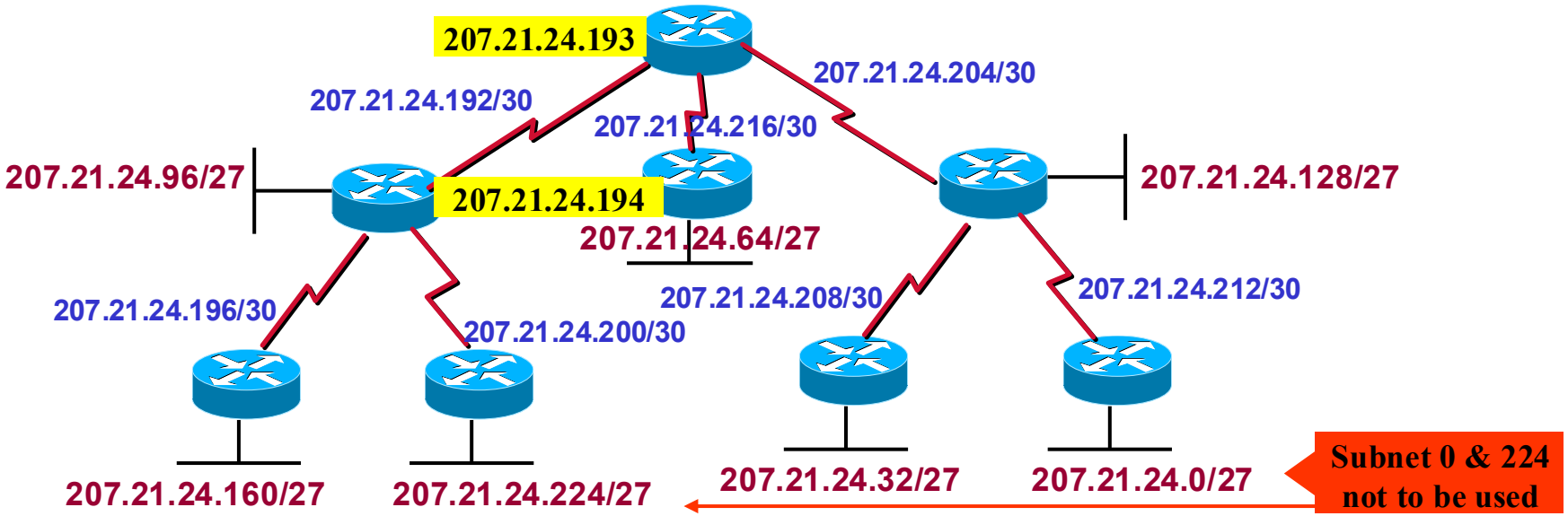
Sub-subnet 0	207.21.24.192	/30
Sub-subnet 1	207.21.24.196	/30
Sub-subnet 2	207.21.24.200	/30
Sub-subnet 3	207.21.24.204	/30
Sub-subnet 4	207.21.24.208	/30
Sub-subnet 5	207.21.24.212	/30
Sub-subnet 6	207.21.24.216	/30
Sub-subnet 7	207.21.24.220	/30

207.21.24.192/27 subnet, subnetted into eight /30 subnets

- This network has seven /27 subnets with 30 hosts each *AND* eight /30 subnets with 2 hosts each (excluding special host addresses)
- /30 subnets are very useful for serial networks
  - exactly 2 host addresses for serial link interfaces

Subnet 0	207.21.24.0	/27
Subnet 1	207.21.24.32	/27
Subnet 2	207.21.24.64	/27
Subnet 3	207.21.24.96	/27
Subnet 4	207.21.24.128	/27
Subnet 5	207.21.24.160	/27
~~Subnet 6~~	~~207.21.24.192~~	~~/27~~
Subnet 7	207.21.24.224	/27

Sub-subnet 0	207.21.24.192	/30
Sub-subnet 1	207.21.24.196	/30
Sub-subnet 2	207.21.24.200	/30
Sub-subnet 3	207.21.24.204	/30
Sub-subnet 4	207.21.24.208	/30
Sub-subnet 5	207.21.24.212	/30
Sub-subnet 6	207.21.24.216	/30
Sub-subnet 7	207.21.24.220	/30



- This network uses all seven /27 subnets with 30 hosts each AND seven out of eight /30 subnets with 2 hosts each
- /30 subnets with 2 hosts per subnet do not waste host addresses

Subnet 0	207.21.24.0	/27
Subnet 1	207.21.24.32	/27
Subnet 2	207.21.24.64	/27
Subnet 3	207.21.24.96	/27
Subnet 4	207.21.24.128	/27
Subnet 5	207.21.24.160	/27
Subnet 6	207.21.24.192	/27
Subnet 7	207.21.24.224	/27

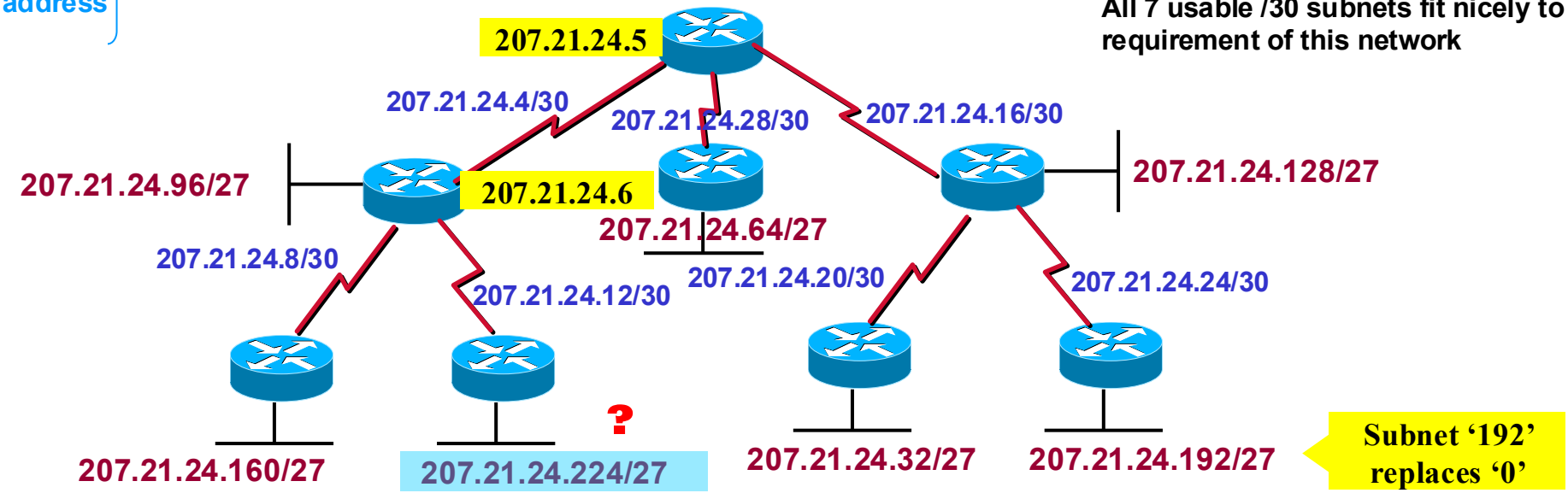
Subnetting upper boundary /27 block address

207.21.24.224/28 √
207.21.24.240/28 ×

Subnetting lower boundary /27 block address

Sub-subnet 0	207.21.24. 0	/30
Sub-subnet 1	207.21.24. 4	/30
Sub-subnet 2	207.21.24. 8	/30
Sub-subnet 3	207.21.24. 12	/30
Sub-subnet 4	207.21.24. 16	/30
Sub-subnet 5	207.21.24. 20	/30
Sub-subnet 6	207.21.24. 24	/30
Sub-subnet 7	207.21.24. 28	/30

All 7 usable /30 subnets fit nicely to the requirement of this network



- This network uses all seven /27 subnets with 30 hosts each AND all seven usable /30 subnets with 2 hosts each.

CIDR (Classless Inter-Domain Routing)

CIDR (Classless Inter-Domain Routing)

- Supernetting or CIDR, “cider” prevents internet routing tables explosion.
- To CIDR-compliant routers, address class is meaningless.
  - No more traditional Class A, B, C in routing.
  - The network portion of the address is determined by the network subnet mask or prefix-length (/8, /19, etc.)
  - The network address in CIDR is NOT determined by the first octet (first two bits), e.g. 200.10.0.0/16 or 15.10.160.0/19
- CIDR routers aggregate, or ***summarize***, routing information and thus shrink the size of their routing tables.
  - Just one address and mask combination can represent the routes to multiple networks.
  - Dynamic routing protocols must send network address (network prefix) and mask (prefix-length) information in their routing updates
  - Route summarization is also known as: route aggregation or supernetting
- Supernetting is essentially the inverse of subnetting

Without CIDR, a router must maintain individual routing table entries for these class B networks.

Network Number	First Octet	Second Octet	Third Octet	Fourth Octet
172.24.0.0 /16	10101100	00011000	00000000	00000000
172.25.0.0 /16	10101100	00011001	00000000	00000000
172.26.0.0 /16	10101100	00011010	00000000	00000000
172.27.0.0 /16	10101100	00011011	00000000	00000000
172.28.0.0 /16	10101100	00011100	00000000	00000000
172.29.0.0 /16	10101100	00011101	00000000	00000000
172.30.0.0 /16	10101100	00011110	00000000	00000000
172.31.0.0 /16	10101100	00011111	00000000	00000000

With CIDR, a router can summarize these eight networks by using a 13-bit prefix: **172.24.0.0 /13**

Network Number	First Octet	Second Octet	Third Octet	Fourth Octet
172.24.0.0 /16	10101100	00011000	00000000	00000000
172.25.0.0 /16	10101100	00011001	00000000	00000000
172.26.0.0 /16	10101100	00011010	00000000	00000000
172.27.0.0 /16	10101100	00011011	00000000	00000000
172.28.0.0 /16	10101100	00011100	00000000	00000000
172.29.0.0 /16	10101100	00011101	00000000	00000000
172.30.0.0 /16	10101100	00011110	00000000	00000000
172.31.0.0 /16	10101100	00011111	00000000	00000000

Steps:

1. Count the number of left-most matching bits, /13
2. Add all zeros after the last matching bit:

172.24.0.0 = 10101100 00011000 00000000 00000000

Supernetting Example

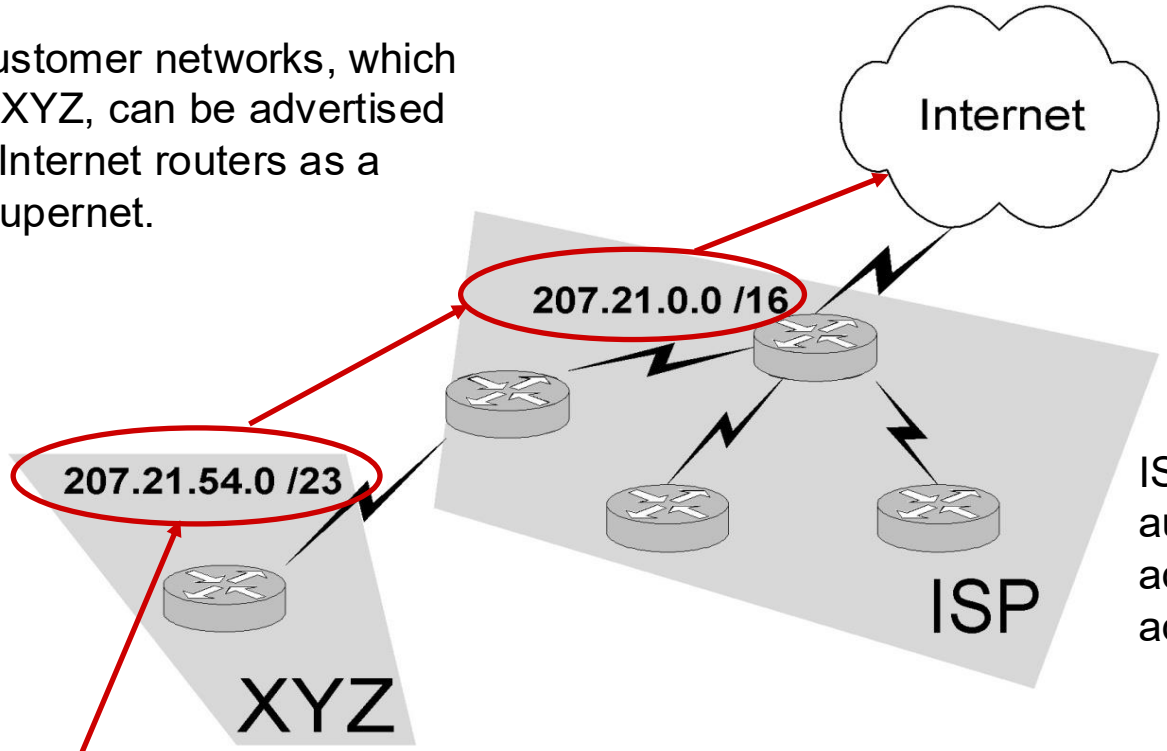
- **Company XYZ needs to address 400 hosts.**
- **Its ISP gives them two contiguous Class C addresses:**
  - **207.21.54.0/24**
  - **207.21.55.0/24**
- **Company XYZ can use a prefix of 207.21.54.0 /23 to supernet these two contiguous networks. (Yielding 508 hosts)**
- **207.21.54.0 /23**
  - **207.21.54.0/24**
  - **207.21.55.0/24**

Network Number	First Octet	Second Octet	Third Octet	Fourth Octet
207.21.54.0	11001111	00010101	00110110	00000000
207.21.55.0	11001111	00010101	00110111	00000000

23 bits in common

Supernetting Example

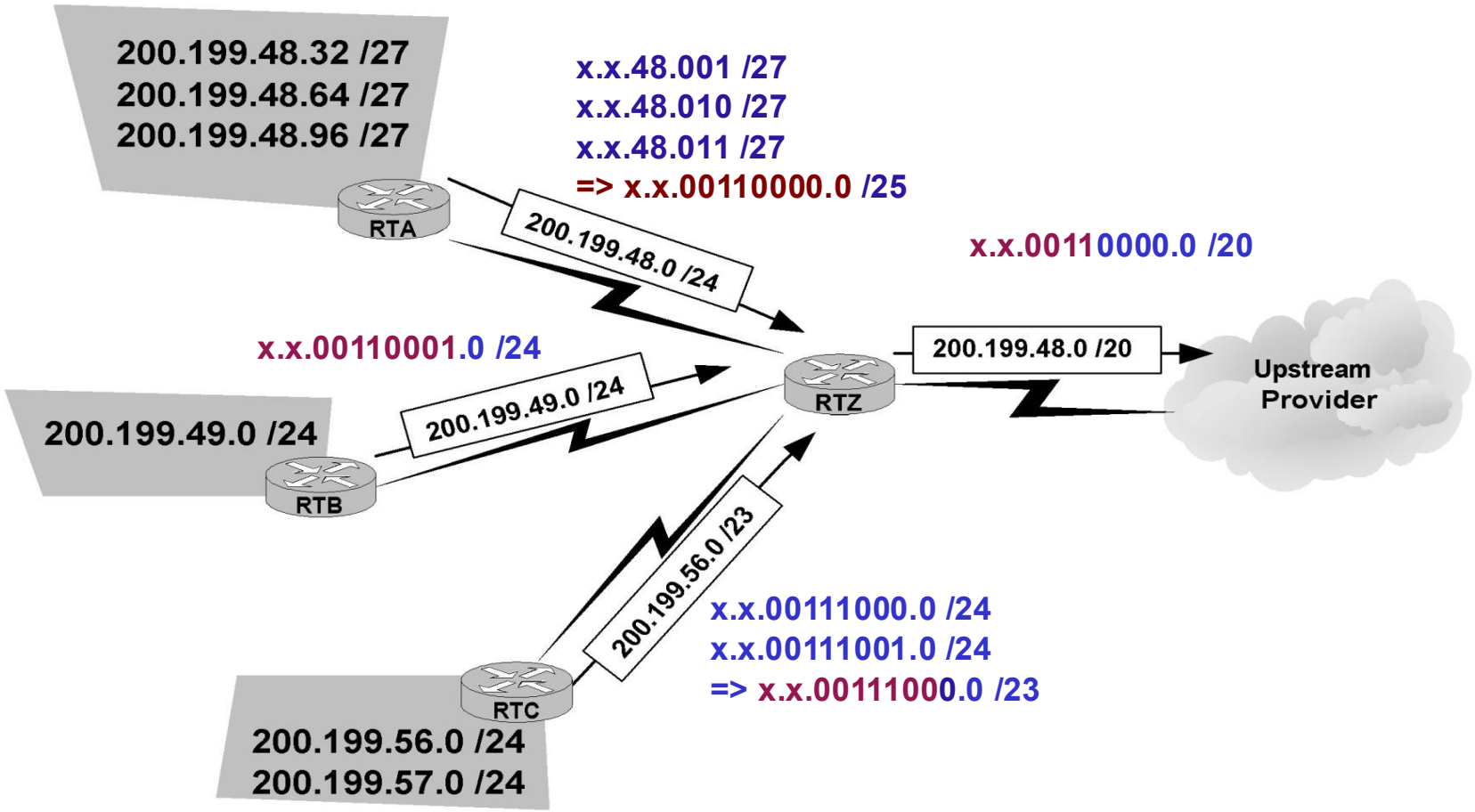
ISP's customer networks, which include XYZ, can be advertised among Internet routers as a single supernet.



ISP acts as the addressing authority for a CIDR block of addresses, assigns supernet address to XYZ

Network Number	First Octet	Second Octet	Third Octet	Fourth Octet
207.21.54.0	11001111	00010101	00110110	00000000
207.21.55.0	11001111	00010101	00110111	00000000

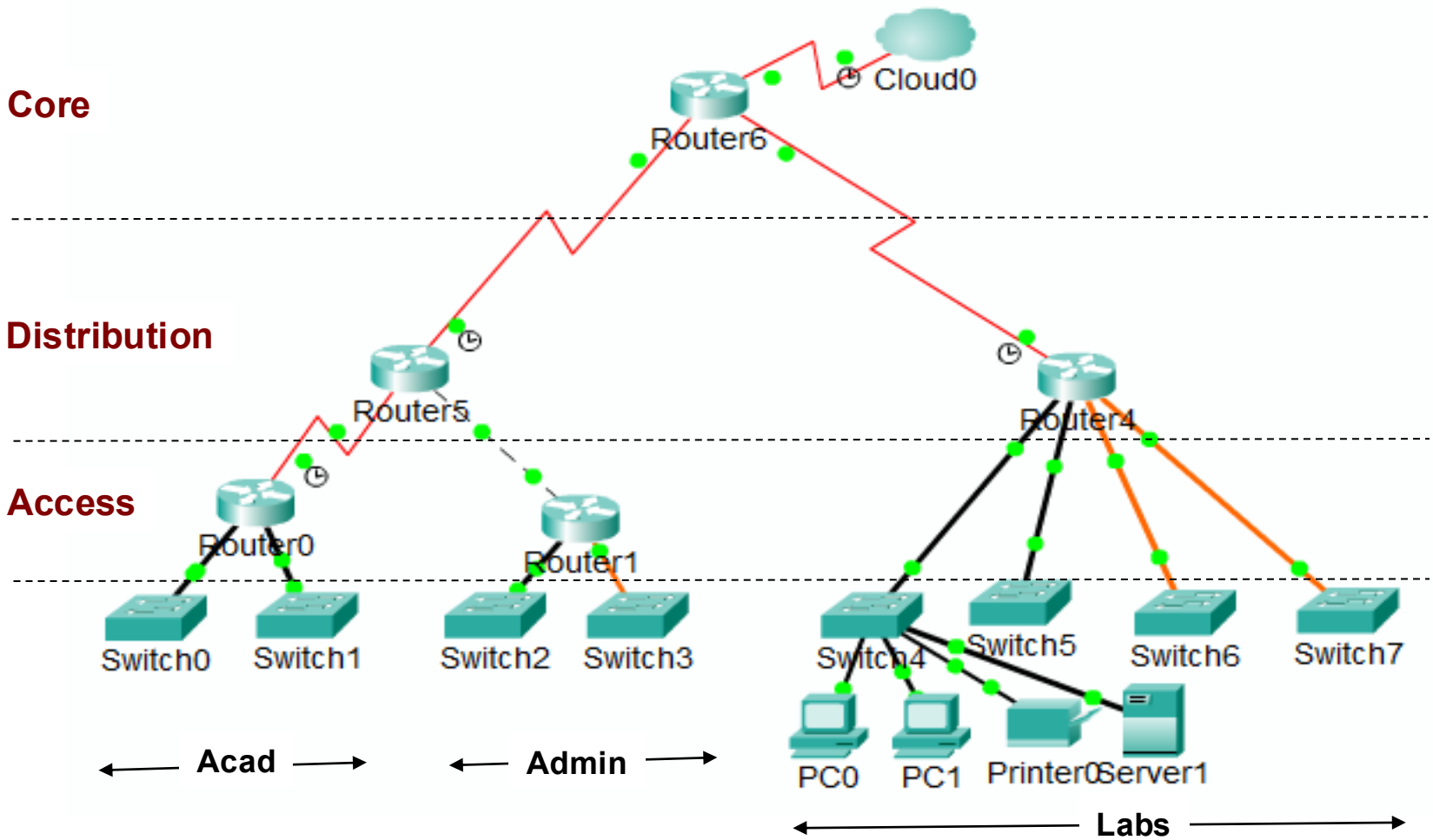
CIDR and the Provider



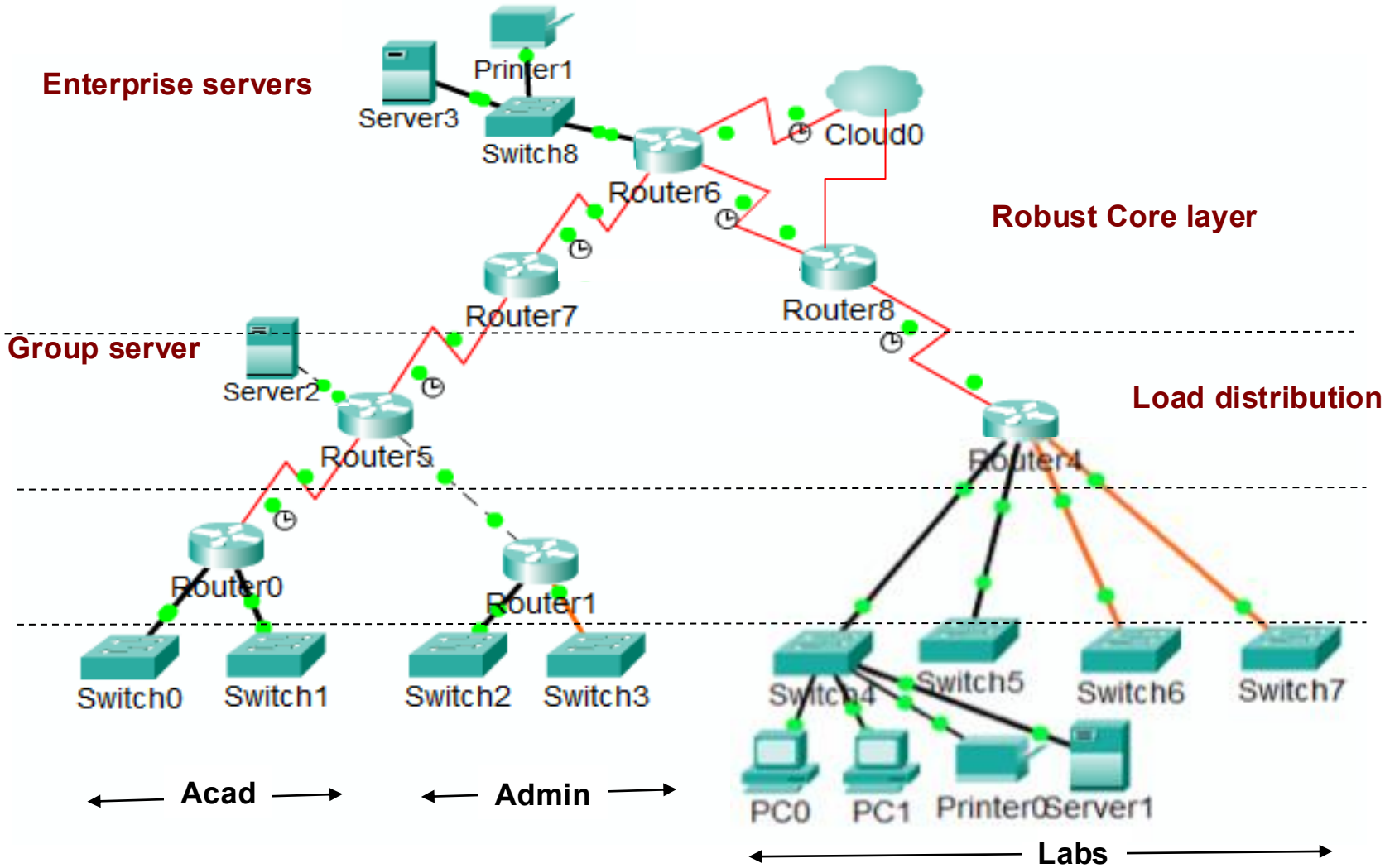
Another example of route aggregation.

Optimizing the Network Design

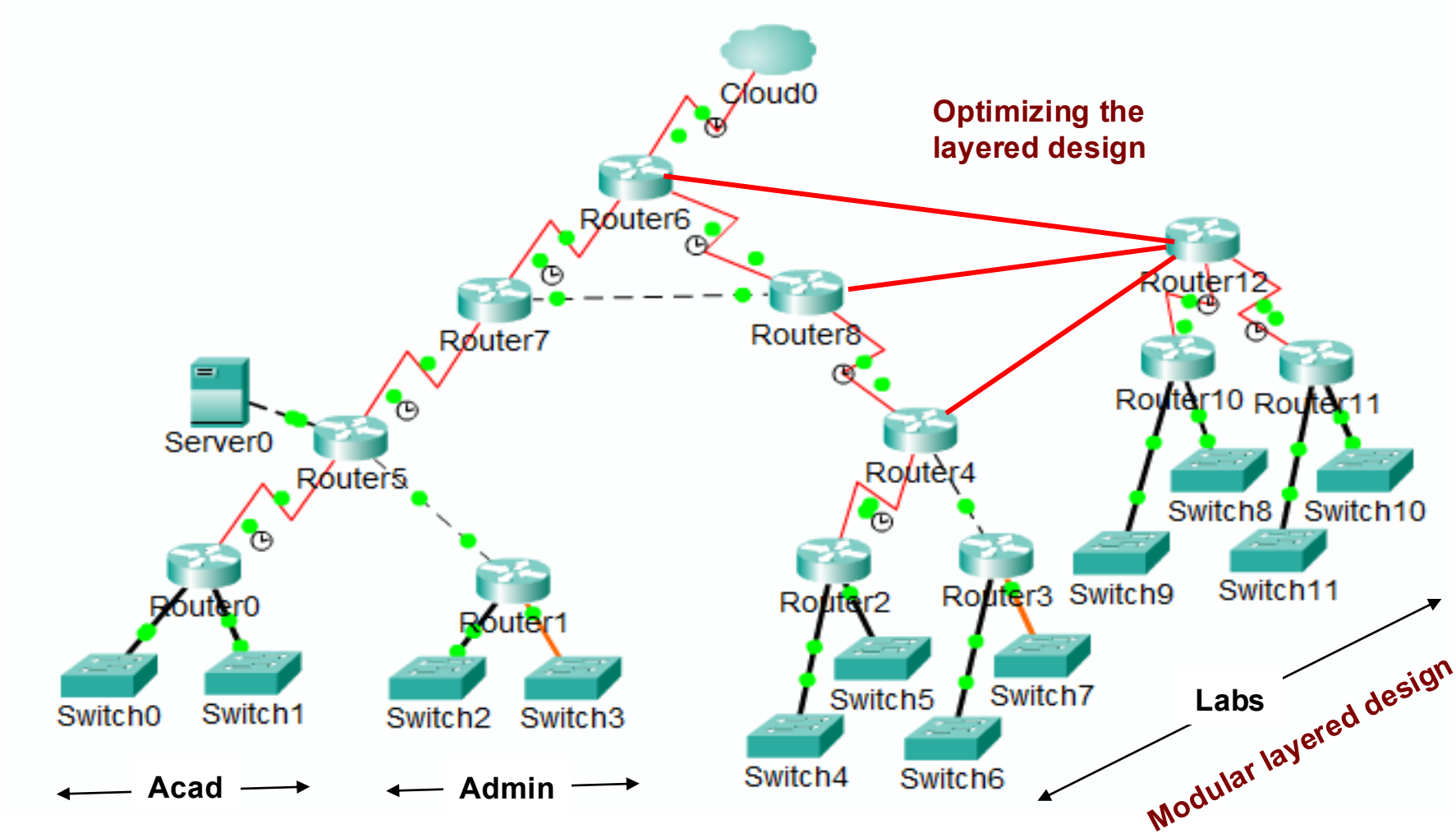
A 3-layered Enterprise Network



Load Distribution & Server Placement

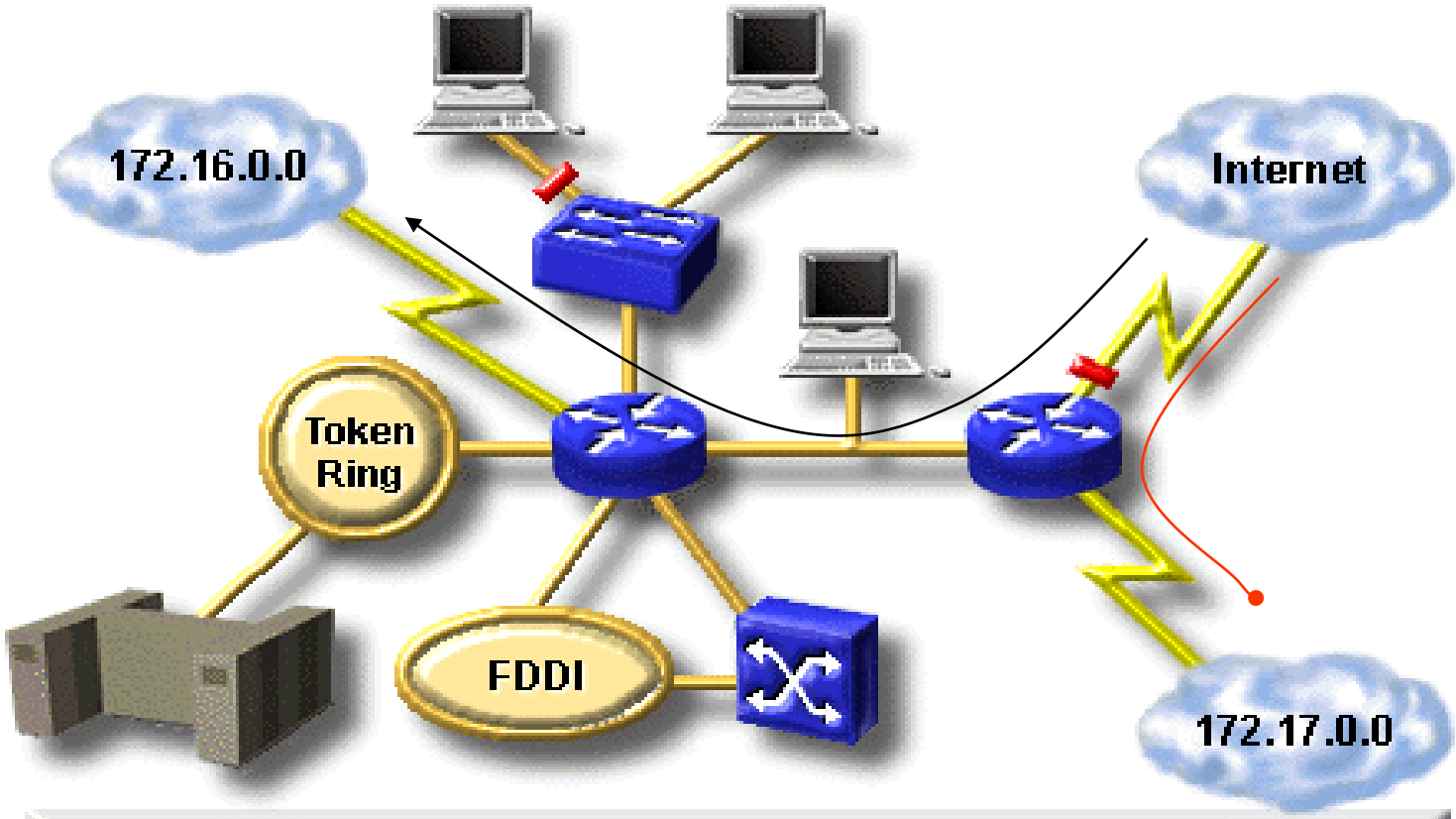


Modular & Layered Design



Access List Overview

Why Use Access Lists?

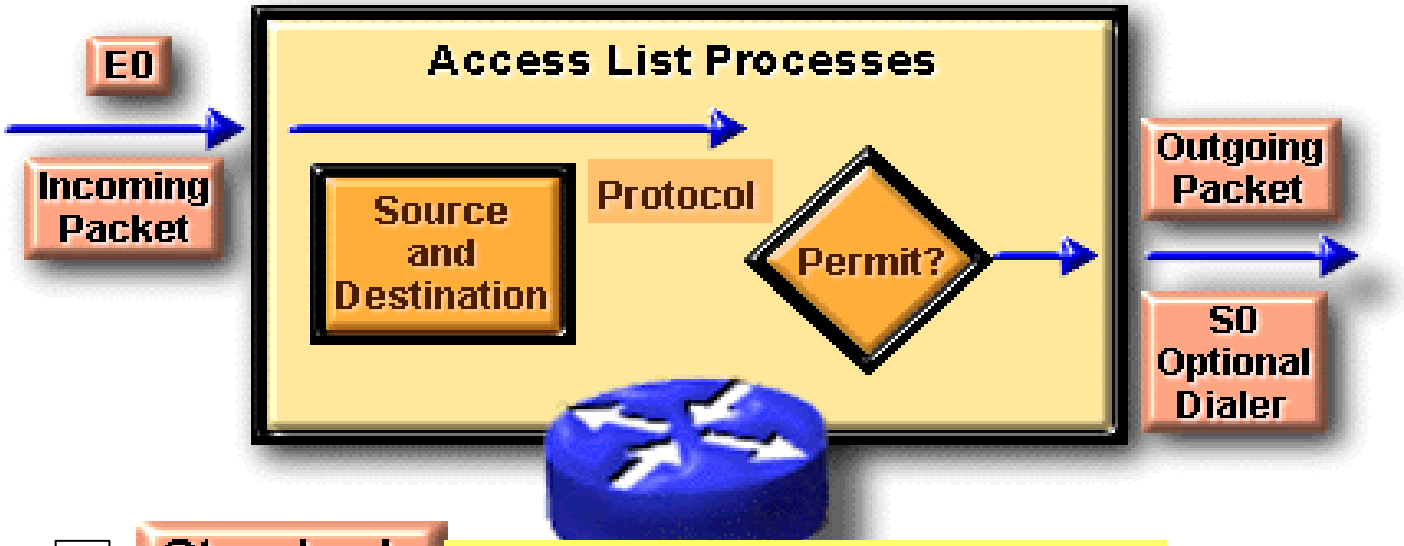


- Deny traffic you do not want based on packet tests (for example, addressing or traffic type)

Access Control Lists - Security Requirements

- User Access Control
  - Lab users, Research students, Administrative staff, Academic staff
  - Host number filtering
- Server Access Control
  - Application sharing and restriction
  - Port number filtering
- Network Access Control
  - Lab, Research, Administrative, and Academic Networks
  - Subnet address filtering

What Are Access Lists?



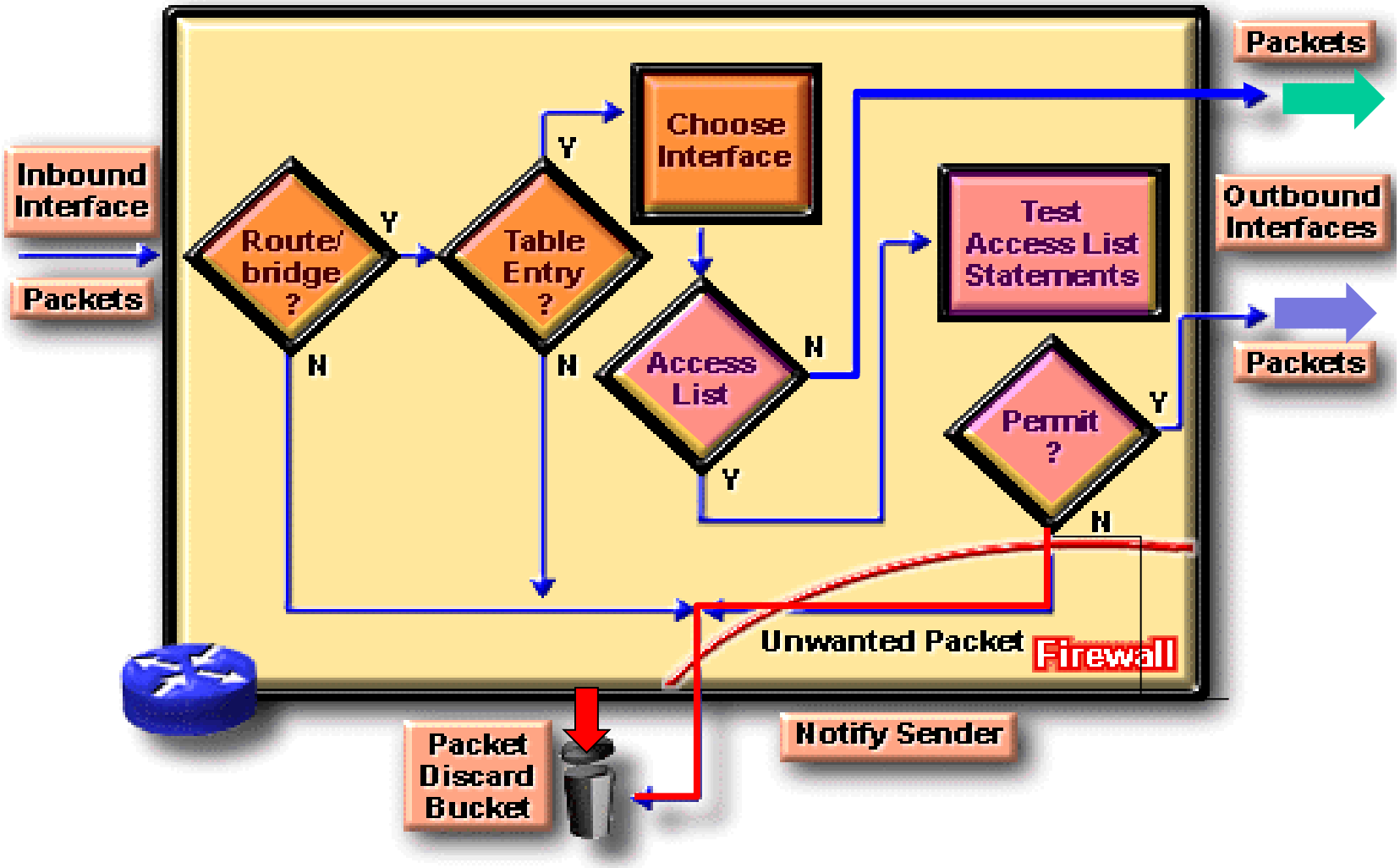
1 **Standard** — Source network/ subnet/ host address

- Simpler address specifications
- Generally permits or denies entire protocol suite

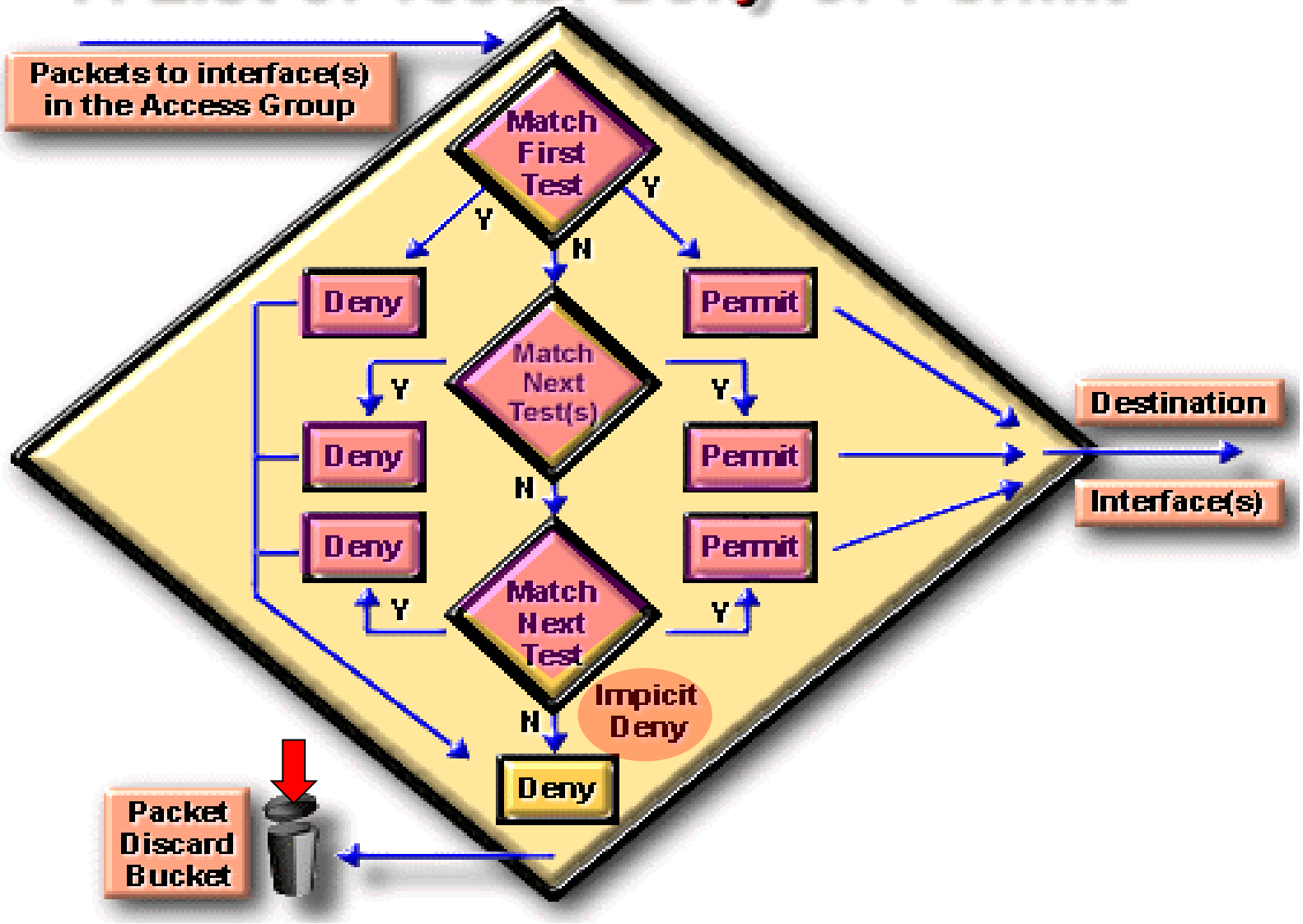
2 **Extended** — Source/ destination addresses, protocols, port numbers, and other parameters

- More complex address specifications
- Generally permits or denies specific protocols

How Access Lists Work



A List of Tests: Deny or Permit



Access List Command Overview

Step 1: Set parameters for this access list test statement (which can be one of several statements)

Router(config)#

```
access-list access-list-number {permit I deny} {test conditions}
```

Step 2: Enable an interface to become part of the group that uses the specified access list

Router(config-if)#

```
{protocol} access-group access-list-number { in | out}
```

- Access lists are numbered (for IP, numbered or named)

There can be only one access list per protocol per interface

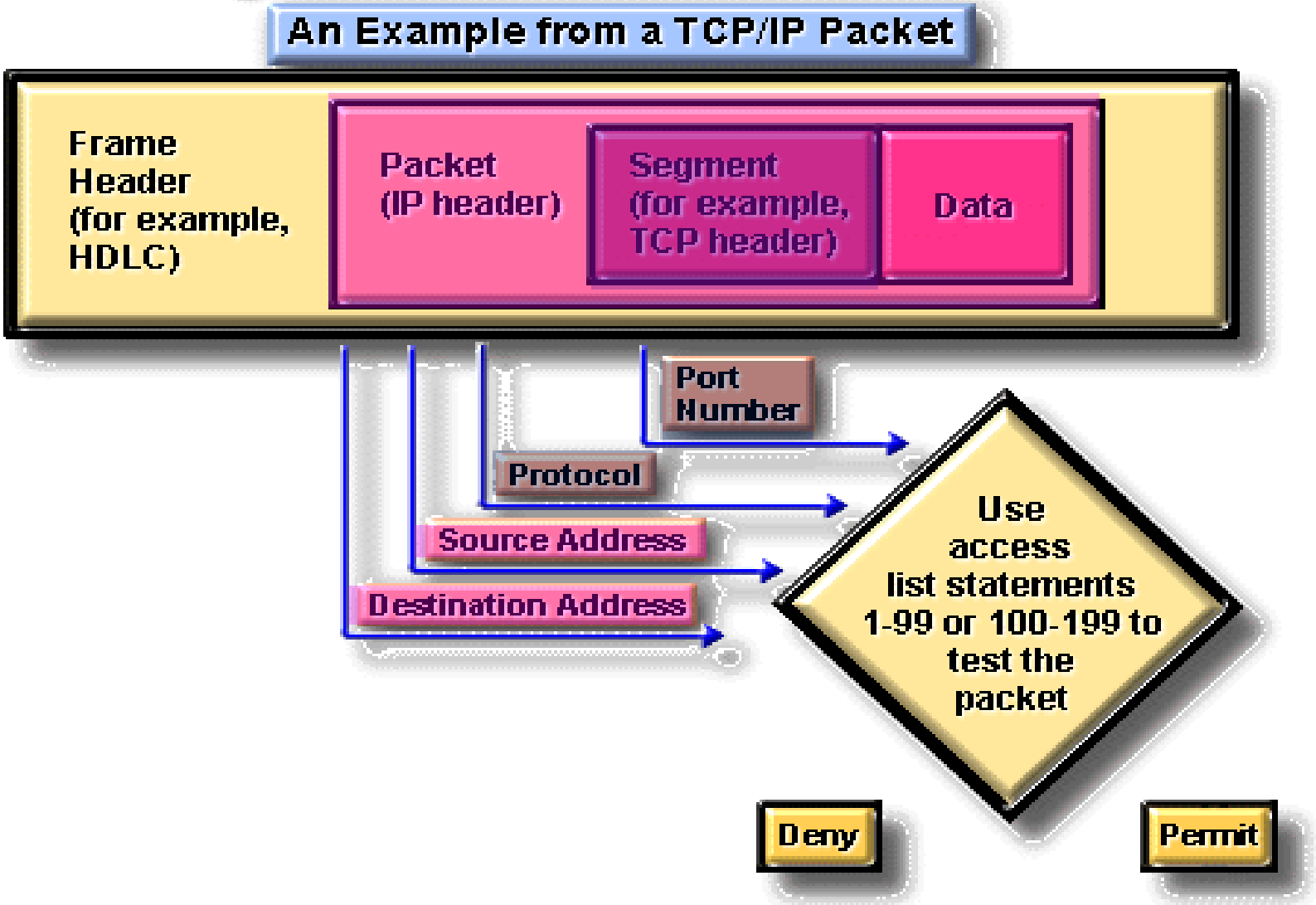
How to Identify Access Lists

Access List Type		Number Range/Identifier
IP	Standard	1-99
	Extended	100-199
		Named (Cisco IOS 11.2 and later)
IPX	Standard	800-899
	SAP filters	1000-1099
Apple Talk		600-699

- Number identifies the protocol and type
- Other number ranges for most protocols

TCP/IP Access Lists

Testing Packets with Access Lists



Key Concepts for IP Access Lists

- **Standard lists (1 to 99) test conditions of all IP packets from source addresses**
- **Extended lists (100 to 199) can test conditions of**
  - Source and destination addresses
  - Specific TCP/IP-suite protocols
  - Destination ports

Wildcard bits indicate how to check the corresponding address bits (0=check, 1=ignore)

How to Use Wildcard Mask Bits

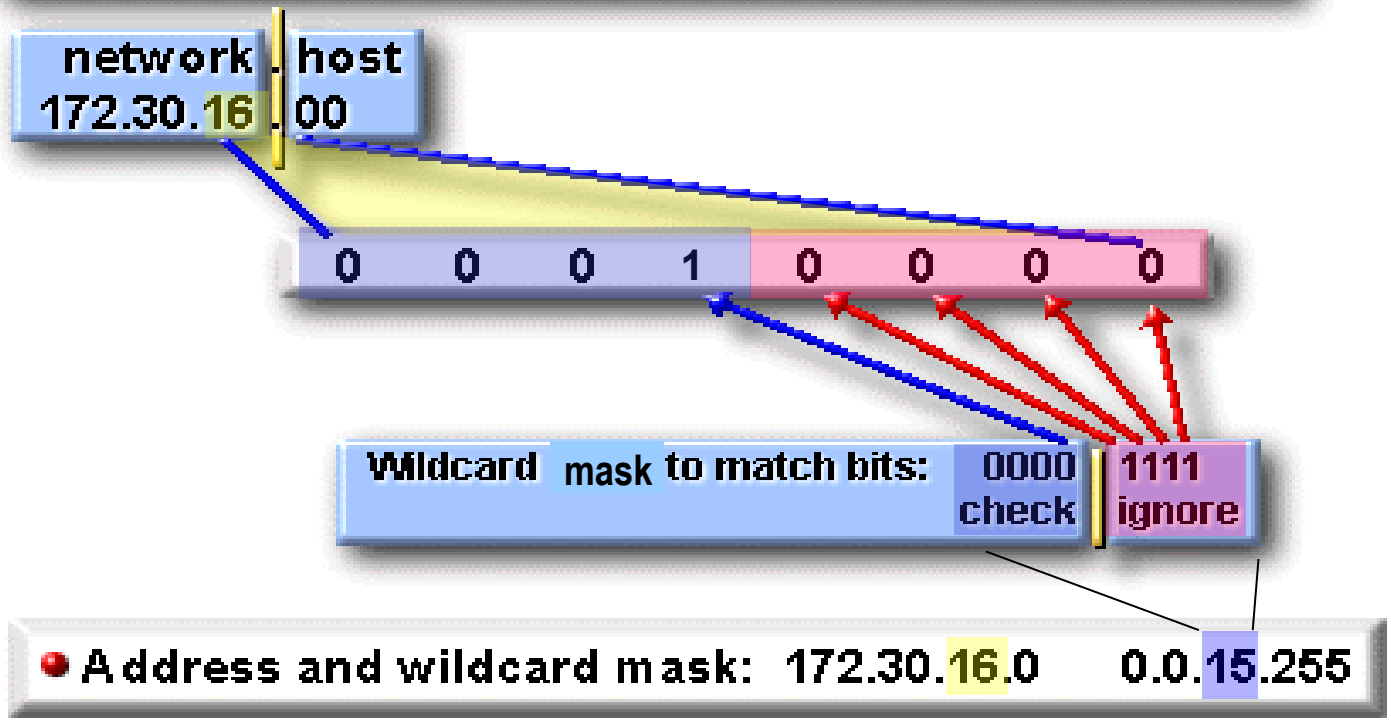
128	64	32	16	8	4	2	1		Octet bit position and address value for bit
0	0	0	0	0	0	0	0	=	check all address bits (match all)
0	0	1	1	1	1	1	1	=	ignore last 6 address bits
0	0	0	0	1	1	1	1	=	ignore last 4 address bits
1	1	1	1	1	1	0	0	=	check last 2 address bits
1	1	1	1	1	1	1	1	=	do not check address (ignore bits in octet)

Examples

- 0 means check corresponding bit value
- 1 means ignore value of corresponding bit

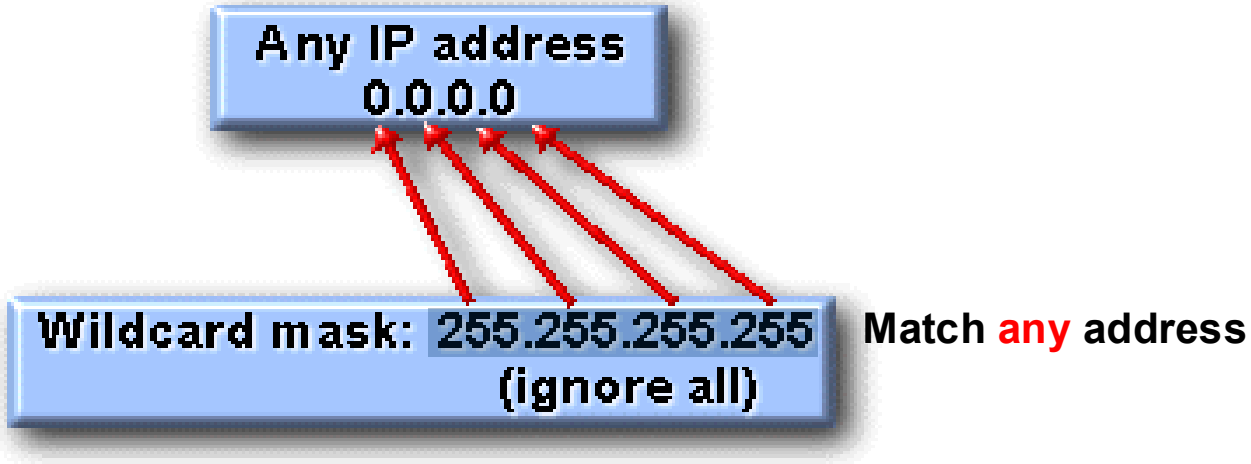
How to Use Wildcard Mask Bits (cont.)

IP access list test conditions:
Check for IP subnets 172.30.16.0 to 172.30.31.0



How to Use the Wildcard *any*

Test conditions: Ignore all the address bits (match any)



- Accept any address: 0.0.0.0 255.255.255.255; abbreviate the expression using the keyword *any*

How to Use the Wildcard *host*

Test conditions: Check all the address bits (match all)

An IP host address, for example:
172.30.16.29

Wildcard mask: 0.0.0.0
(check all bits)

Match a specific address

- Example 172.30.16.29 0.0.0.0 check all the address bits
- Abbreviate the wildcard using the word keyword host, followed by the IP address.
  Ex: host 172.30.16.29

IP Standard Access Configuration

Step 1:

Create access-list on router

Router(config)#

```
access-list access-list-number {permit | deny}
source [source-mask]
```

- Sets parameters for this list entry
- IP standard access lists uses 1 to 99

"access-list" creates a set of filter statements for a particular access-list-number

Step 2:

Enable access control on interface

To select interface: >interface GigabitEthernet 1/0

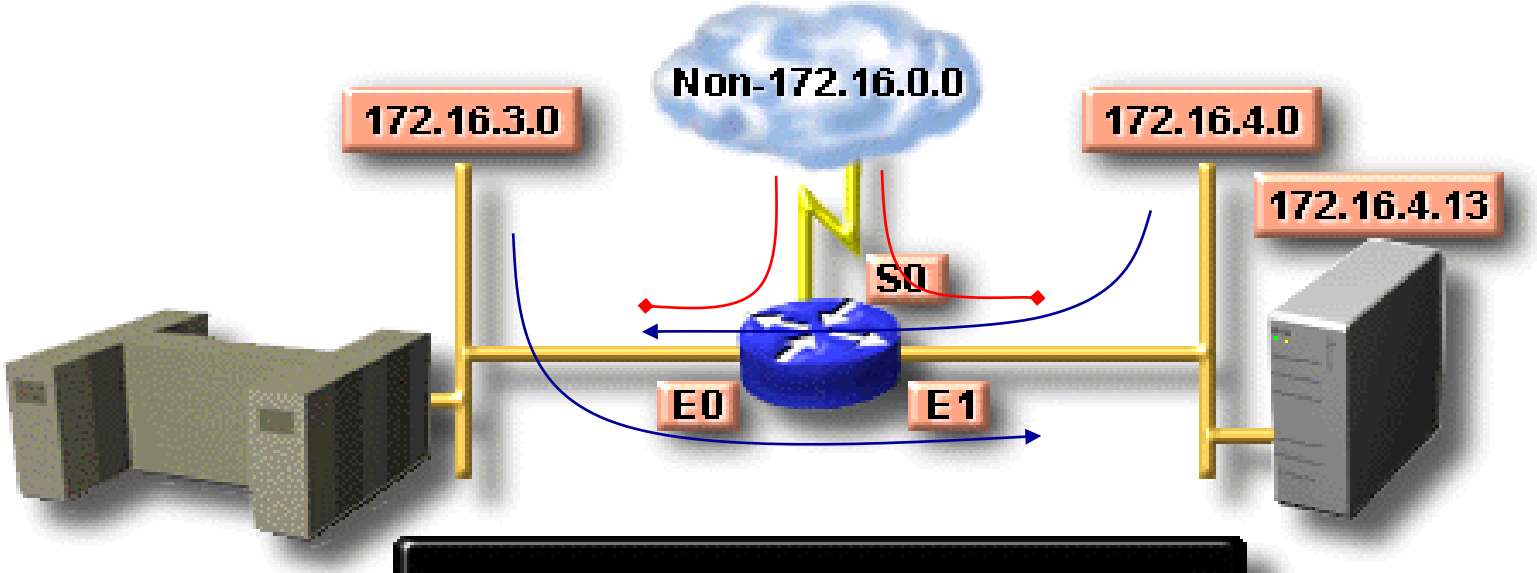
Router(config-if)#

```
ip access-group access-list-number {in | out}
```

- Activates the list on an interface

"ip access-group" associates an existing access-list-number to an interface

Standard Access List Example 1

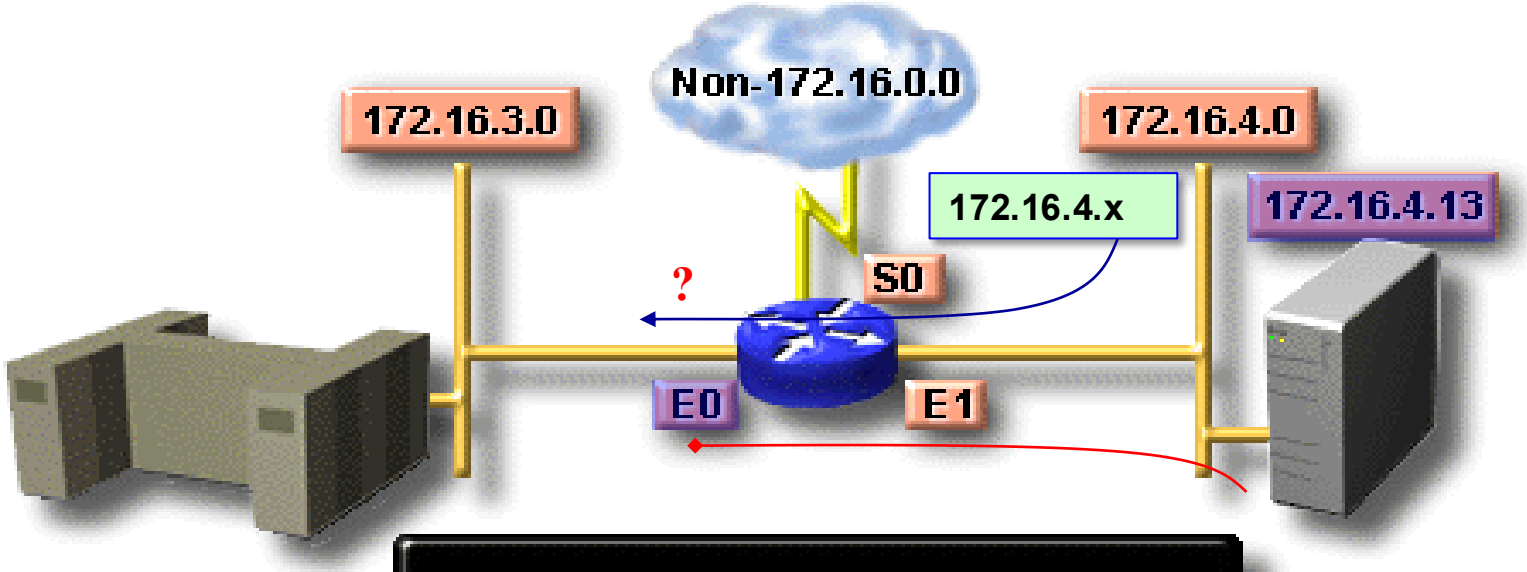


Step 1:

Step 2:

- Permit a specific network

Standard Access List Example 2

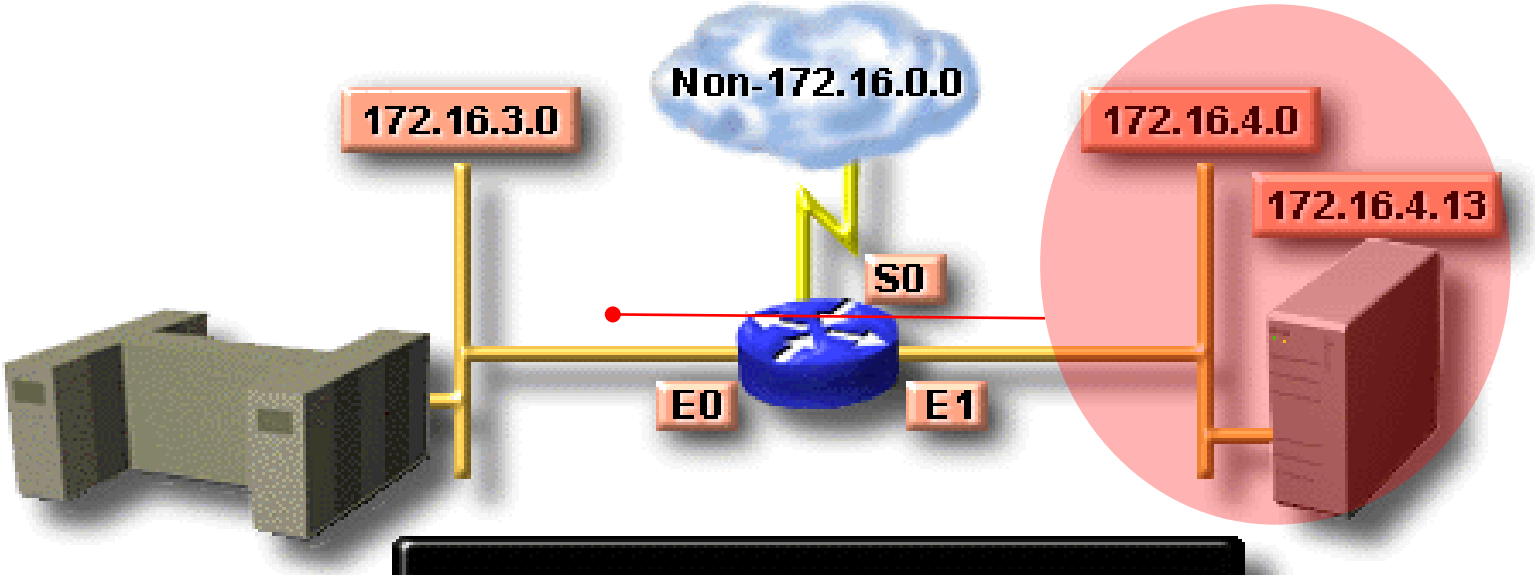


Step 1:

Step 2:

- Deny a specific host

Standard Access List Example 3



Step 1:

Step 2:

- Deny a specific subnet

Extended IP Access Lists

Allow more precise filtering conditions

- Check source and destination IP address
- Specify an optional IP protocol port number
- Use access list number range 100 to 199

Extended Access List Configuration

Step 1:

Create access-list on router

Router(config)#

access-list *access-list-number* {permit | deny} protocol *source source-mask destination destination-mask* [*operator operand*] [established]

- Sets parameters for this list entry
- IP uses a list number in range 100 to 199

Operator operand: lt, gt, eq, neq (less than, greater than, equal, not equal), and a port number.

Step 2:

Enable access control on interface

ip access-group *access-list-number* {in | out}

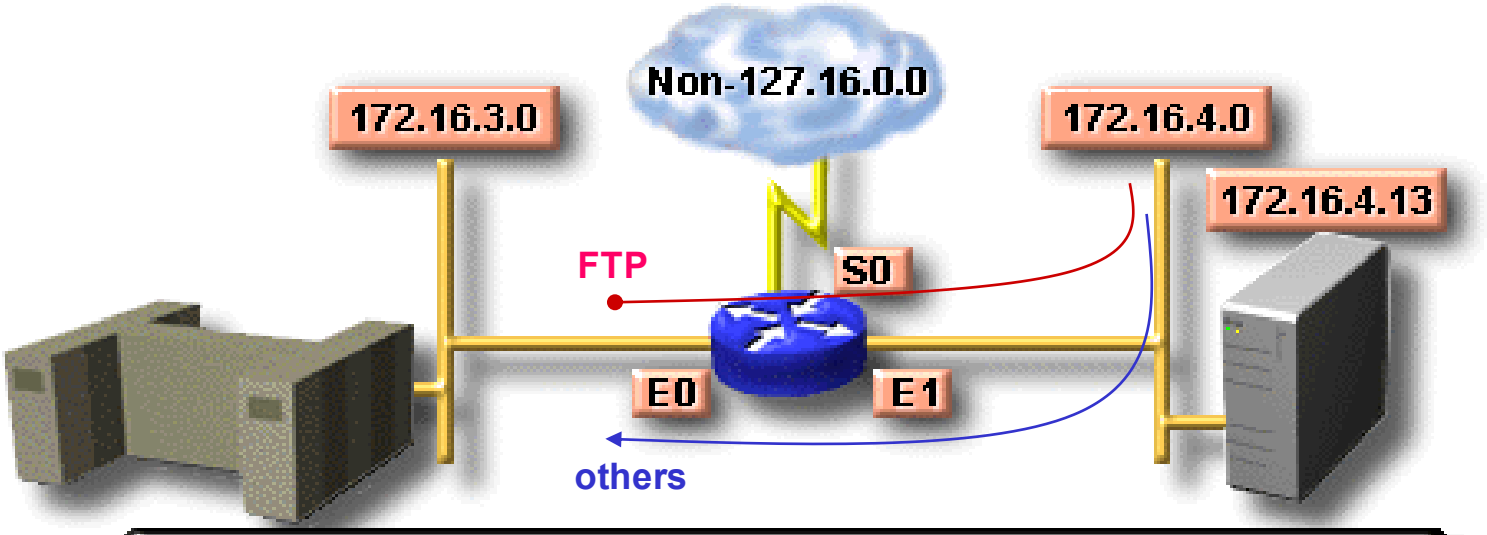
- Activates the extended list on an interface

Well-Known Port Numbers

Well-Known Port Number (Decimal)

- 20 File Transfer Protocol (FTP) data
- 21 FTP program control
- 23 Telnet
- 25 Simple Mail Transport Protocol (SMTP)
- 69 Trivial File Transfer Protocol (TFTP)
- 53 Domain Name System (DNS)
- 80 World Wide Web (HTTP)
- 443 Secured World Wide Web (HTTPS)

Extended Access List Example 1



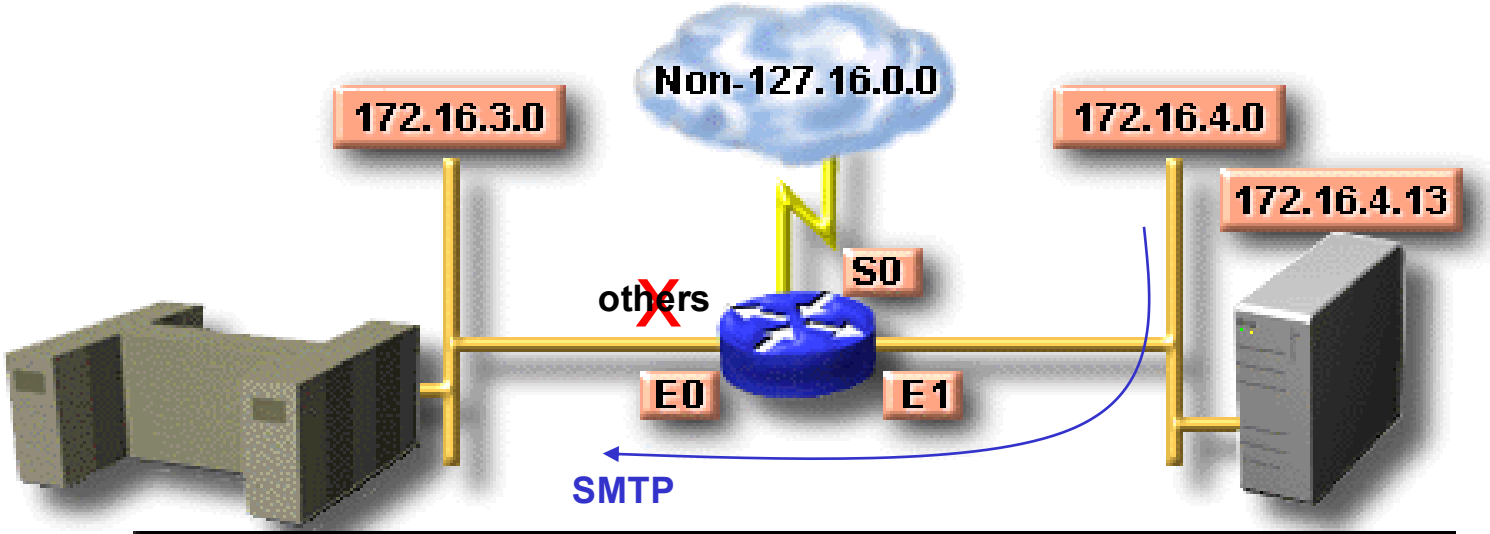

```
access-list 101 deny tcp 172.16.4.0 0.0.0.255 172.16.3.0 0.0.0.255 eq 21

(implicit deny any)
(access-list 101 deny ip 0.0.0.0 255.255.255.255 0.0.0.0 255.255.255.255)
```

Allow all other traffic on the same connection

- Deny FTP for E0

Extended Access List Example 2



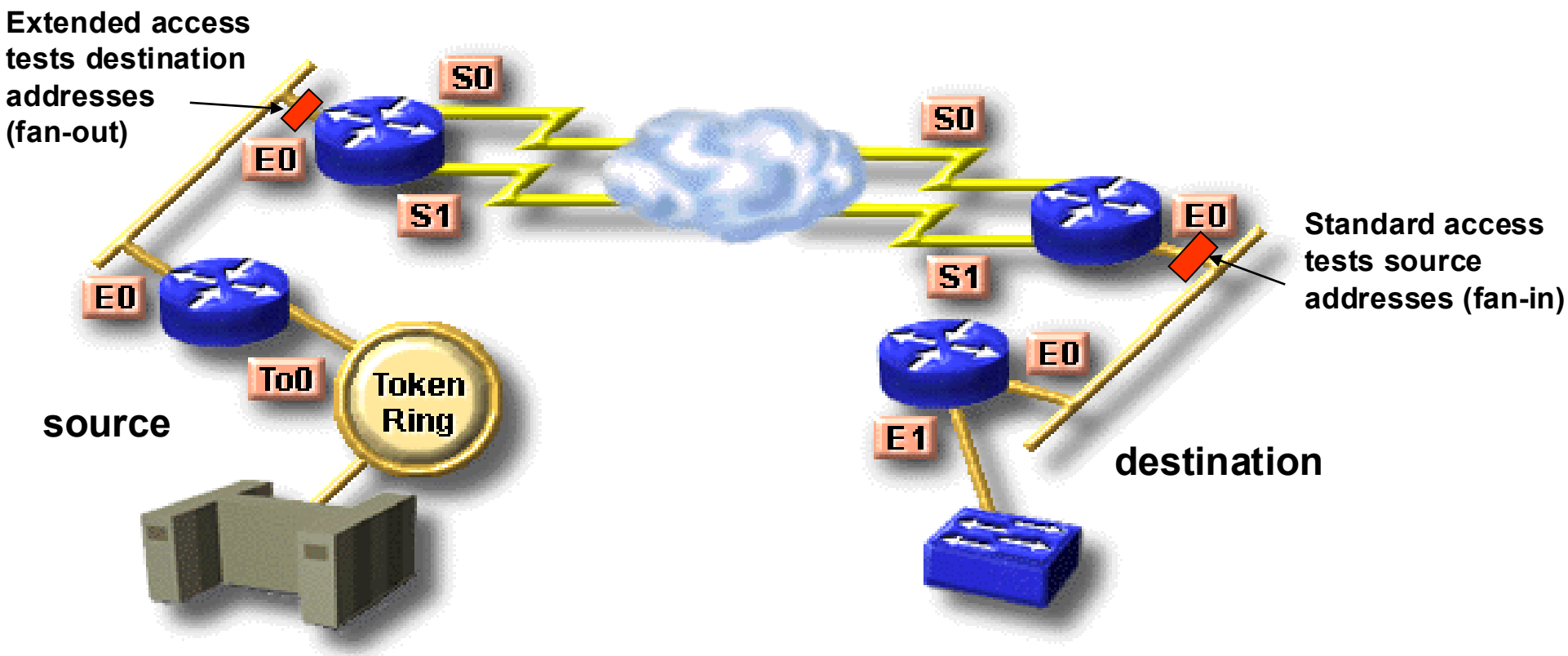

```
access-list 101 permit tcp 172.16.4.0  0.0.0.255  any eq 25
(implicit deny any)
(access-list 101 deny ip 0.0.0.0  255.255.255.255  0.0.0.0  255.255.255.255)

interface ethernet 0
ip access-group 101
```

This example allows only mail from 172.16.4.0 to be sent out at interface E0. All other traffic from any other sources are denied.

- Allow only SMTP for E0

Where to Place IP Access Lists



- Place standard access lists close to the destination
- Place extended access lists close to the source